

MODULE - IIITHEORIES OF INTERNATIONAL BUSINESS**Module - III****(7 Hours)**

Theories of International Business: Introduction, Mercantilism, Theory of absolute cost advantage, Comparative cost advantage theory, Comparative cost advantage with money, Relative factor endowment theory, Product life cycle theory, Global strategic rivalry theory, Porter's National Competitive Advantage Theory.

INTRODUCTION

A number of theories have been developed by the international economists to explain how does international business take place and how do countries get benefit out of international business.

What is International Trade?

The reason for the emergence of international trade is that the human wants are varied and unlimited and no single country possesses the adequate resources to satisfy all these wants. Hence there arises a need for interdependence between countries in the form of international trade. So in order to make effective utilization of the world's resources international trade is to be boosted and the problems faced by the countries should be dealt with.

International trade theories are simply different theories to explain international trade. Trade is the concept of exchanging goods and services between two people or entities. *International trade* is then the concept of this exchange between people or entities in two different countries.

People or entities trade because they believe that they benefit from the exchange. They may need or want the goods or services. While at the surface, this many sound very simple, there is a great deal of theory, policy, and business strategy that constitutes international trade.

Basis of International Trade

No country is self-sufficient in producing all the required goods and services from its own resources. This problem can be solved through international trade where the countries obtain those goods which it cannot produce or cannot produce as cheaply as possible in another country. However this is not the only basis for doing international trade, there are other reasons also. Trade economists have laid down different theories for international trade.

Theories of international trade

Theories of international trade tend to explain the nature and movement of international trade.

Such theories can be classified into:

Classical Country-Based Theories: Mercantilism, Absolute Advantage, Comparative Advantage and Heckher-Ohlin Theory.

Modern Firm-Based Theories: Country Similarity, Product Life Cycle, Global Strategic Rivalry and Porter's National Competitive Advantage.

Classical Country-Based Theories	Modern Firm-Based Theories
Mercantilism Absolute Advantage Comparative Advantage Heckscher-Ohlin	Country Similarity Product Life Cycle Global strategic Rivalry Porter's National Competitive Advantage

These theories include

1. Mercantilism
2. Theory of Absolute Cost Advantage
3. Comparative Cost Advantage Theory
4. Comparative Cost Advantage with Money
5. Relative Factor Endowments/Heckscher-Ohlin Theory
6. Country Similarity Theory
7. Product Life Cycle Theory
8. Global Strategic Rivalry Theory
9. Porter's National Competitive Advantage

A.

CLASSICAL COUNTRY-BASED THEORIES

I.

MERCANTILISM

Mercantilism is the oldest international trade theory that formed the foundation of economic thought during about 1500 to 1800. According to this theory, the holdings of a country's treasure primarily in the form of gold constituted its wealth. This theory specifies that countries should export more than they import and receive the value of trade surplus in the form of gold from those countries which experience trade deficits.

Governments imposed restrictions on imports and encouraged exports in order to prevent trade deficit and experience trade surplus. **Colonial powers like the British used to trade with their colonies like India, Sri Lanka, etc., by importing the raw materials from and exporting the finished goods to colonies. The colonies had to export less valued goods and import more valued goods. Thus, colonies were prevented from manufacturing.** This practice allowed the colonial powers to enjoy trade surplus and forced the colonies to experience trade deficits. The theory benefited the colonial powers and caused much discontent in the colonies. In fact, this was the background situation for the American Revolution.

The Mercantilism theory suggests for maintaining favourable balance of trade in the form of import of gold for export of goods and services. But, the decay of gold standard reduced the validity of this theory consequently; this theory was modified in neo-mercantilism.

Neo-mercantilism proposes that countries attempt to produce more than the demand in the domestic country in order to achieve a social objective like full employment in the domestic country or a political objective like assisting a friendly country.

This theory was attacked on the ground that the wealth of a nation is based on its available goods and services rather than on gold. Adam Smith developed the theory of absolute cost advantage which says

that different countries can get the advantage of international trade by producing certain goods more efficiently than others.

II. THE THEORY OF ABSOLUTE COST ADVANTAGE

Adam Smith, the Scottish economist viewed that mercantilism weakens a country. He advocated free trade among countries to increase a country's wealth. Free trade enables a country to provide a variety of goods and services to its people by specialising in the production theory-of-absolute-cost of some goods and services and importing others. Which goods should a country produce and which goods it should import? Adam Smith proposed a theory to answer this question.

Adam Smith proposed Absolute Cost Advantage Theory of International Trade (1776) based on the principle of division of labour. According to him, application of this principle to international scenario helps the countries to specialise in the production of those goods in which they have cost advantage over other countries.

According to Adam Smith, every country should specialise in producing those products which it can produce at less cost than that of other countries and exchange these products with other products produced cheaply by other countries. Trade between two countries takes place when one country produces one product at less cost than that of another country and the other country has an absolute cost advantage over the first country in producing in any other product.

Skilled Labour and Specialisation Advantage: Countries have absolute cost advantage due to the following reasons

- Suitability of the skill of the labour of the country in producing certain products.
- Specialisation of labour in producing certain products leads to higher productivity and less labour cost per unit of output.
- Economies of scale would reduce the labour cost per unit of output.

1. Natural Advantage: In addition to the skilled labour and specialisation advantage countries do also have natural advantage in producing certain products due to climatic conditions, access to certain resources, etc. **For example, Indian climate suits the production of sweet mangoes, coconut, cotton and cashew nuts.. Sri Lankan climate suits the production of tea, rubber. USA climate supports the production of wheat.** Countries with natural advantage can produce specific products at low cost.

2. Acquired Advantage: In addition to the skilled labour and natural advantages, countries also acquire advantages through technology and skill development. **Japan acquired advantage in steel production through the imports of both iron and coal The reason for this success is that Japan acquired labour saving and material saving technology. Denmark exports silver tableware due to the ability of Danish companies in developing distinctive products.**

Technologically advanced countries acquired abilities to develop substitute products for a number of natural products.

Thus, countries have absolute advantage in producing certain products as discussed above.

For example, England had the absolute advantage in producing textiles, whereas France had the absolute advantage in producing wine. Similarly, India has the absolute advantage in producing pens and Japan has the absolute advantage in producing audio tape recorders.

Assumptions of the Theory:

- Trade is between two countries.
- Only two commodities are traded.
- Free trade exists between the countries.
- The only element of cost of production is labour.

The absolute advantage using two countries and two products. In this example, the countries are India and Japan and the commodities are pens and audio tape recorders. We treat the cost of production in terms of labour input.

Table shows the output of two goods per one day of labour for the two countries.

The Theory of Absolute Cost Advantage: An Example		
Output per one day of Labour		
	Japan	India
Pens	20	60
Audio tape recorders	6	2

Ability of labour to produce different goods/services in a day is known as production possibilities. In Japan, one day on labour can produce either 20 pens or 6 audio tape recorders. In India, one day of labour can produce either 60 pens or audio tape recorders. Japan has an absolute advantage in the production of audio tape recorders and India has an absolute advantage in the production of pens. One day of labour in India produces 60 pens, whereas only 20 pens in Japan. It is clear that Japan has absolute advantage in producing audio tape recorders and India in producing pens.

Assume that India and Japan are able to trade with one another, and then both will get the advantage. Suppose Japan agree to exchange four audio tape recorders for 40 pens. Two days of Japanese labour is needed to produce 40 pens and only 0.67 day of Japanese labour is enough to produce 4 audio tape recorders. Thus, Japan can save 1.33 ($2 - 0.67 = 1.33$) days of labour, if it exports audio tape recorders to India and imports pens from India.

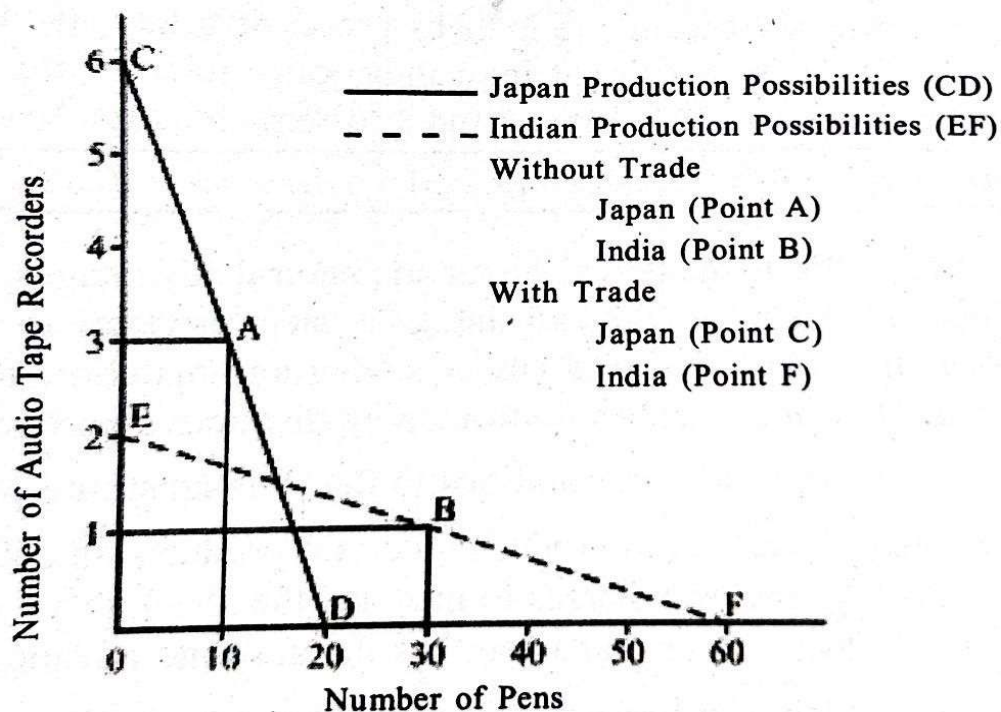
India needs two days of labour to produce 4 audio tape recorders and 0.67 day of labour is enough to produce 40 pens. India can also save 1.33 ($2 - 0.67 = 1.33$) days of labour by exporting pens to Japan and importing audio tape recorders from Japan.

Thus, two countries save labour by trading with each other rather than by producing both the products. The labour can be used for the production of more audio tape recorders by Japan and for the production of more pens by India. Japan can consume more pens by allocating its labour to produce audio tape recorders and by trading with India. And the vice versa is true in case of India.

Japan and India need both audio tape recorders and pens. Japan can produce 10 pens and 3 audio tape recorders with one day's labour without trade. Similarly, India can produce 30 pens and 1 tape recorder with one day's of labour without trade. If Japan and India agree to trade with each other, then Japan can concentrate in producing audio tape recorders and India can concentrate in producing pens.

Similarly, if these two countries agree to exchange 10 pens for 1 tape recorder, then Japan can exchange 3 audio tape recorders for 30 pens. In other words, India gets 3 audio tape recorders for 30 pens. Thus, Japan can enjoy 20 more pens ($30 - 10 = 20$) in addition to 3 audio tape recorders and India can enjoy 2 more audio tape recorders ($3 - 1 = 2$) in addition to 30 pens. Thus, two countries can get the advantage by trading with each other due to absolute cost advantage they possess.

It is clear that both the countries can enjoy 3 audio tape recorders and 30 pens per one day of labour. It does mean that Japan can have the advantage of ($30 - 10 = 20$) 20 pens and India can have the advantage of ($3 - 1 = 2$) 2 audio tape recorders.



Thus, both the countries are better off by producing only one product in which it has absolute cost advantage and trade with the another country for the second product.

Implications of Absolute Cost Advantage Theory

- By trading, two countries can have more quantities of both the products.
- Living standards of the people of both the countries can be increased by trading between the countries.
- Inefficiency in producing certain products in some countries can be avoided.
- Global efficiency and effectiveness can be increased by trading.
- Global labour productivity and other resources productivity can be maximised.

Criticism of Absolute Cost Advantage Theory

1. No Absolute Advantage: According to this theory, one country should be able to produce at least one product at a comparatively low cost. But, in reality, most of the developing countries do not have absolute advantage of producing any product at the lowest cost. Yet, they participate in international trade.

2. Country Size: Countries vary in size. This theory does not deal with country by country differences in specialisation.

3. Variety of Resources: Though there are several resources like labour, technology and natural resources, this theory deals with only labour and ignores all other resources.

4. Transport Cost: Though the cost of transportation plays a significant role in international trade, this theory ignored this aspect.

5. Scale Economies: Large-scale economies reduce the cost of production and form a part of the absolute advantage. But, this theory ignored that aspect also.

6. Absolute Advantage for Many Products: Some countries may have absolute advantage for many products.

For example: Japan, the USA, France, UK, etc. But, this theory does not deal with such situations.

These deficiencies in the absolute cost advantage theory led to the development of other theories. Important among them is comparative cost advantage theory proposed by David Ricardo.

III. COMPARATIVE COST ADVANTAGE THEORY

Absolute cost Advantage Theory fails to explain the situation when one country has absolute cost advantage in producing many products. David Ricardo, a British economist expanded the absolute cost advantage theory to clarify this situation and developed the Theory of Comparative Cost Advantage.

Assumptions of the Theory: The assumptions of the comparative cost advantage theory include:

- There exists full employment.
- The only element of cost of production is labour. Production is subject to the law of constant returns.
- There are no trade barriers,
- Trade is free from cost of production.
- Trade takes place only between two countries.
- Only two products are traded.
- There are no costs of transport, etc.

Comparative cost advantage theory states that a country should produce and export those products for which it is relatively more productive than that of other countries and import those goods for which other countries are relatively more productive than it is

The comparative cost advantage theory is based on relative productivity differences and incorporates the concept of opportunity cost.

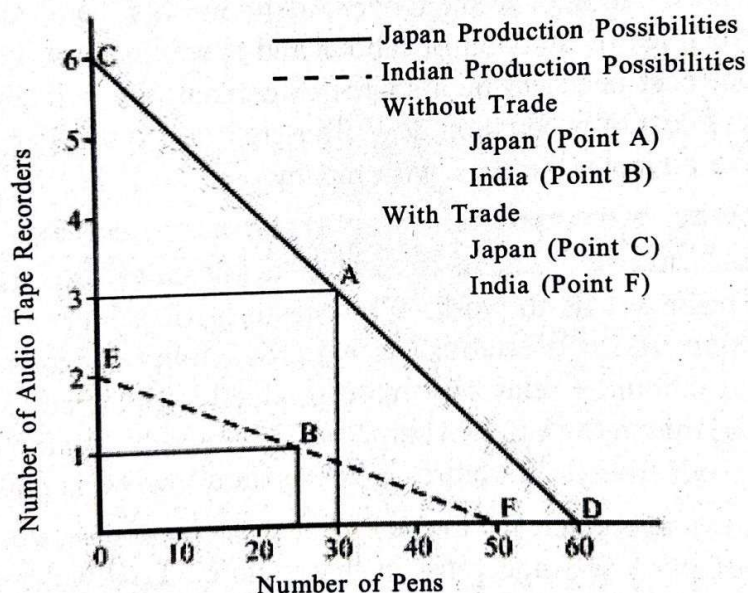
The Theory of Comparative Cost Advantage: An Example		
Output per one day of Labour		
	Japan	India
Pens	60	50
Audio tape recorders	6	2

Japan now produces 60 pens or six audio tape recorders per day of labour. Japan now has an absolute cost advantage in both pens and audio tape recorders. For one day of labour, Japan can produce 10 more pens ($60-50=10$) or 4 more audio tape words ($6-2=4$) than India. Japan is more productive than India in both the products. As much trade should take place between Japan and India based on absolute cost advantage theory

Japan is 3 times better than India in audio tape recorder production, and 1.2 times better in pen production. Alternatively, India is only 0.33 as good as Japan in audio tape recorder production, but 0.83 as good in pen production. Thus comparatively Japan is better in audio tape recorder production and India is better in pens production.

In the absence of trade between two countries, the labour of Japan can produce either 60 pens or 6 audio tape recorder or alternatively 30 pens (50 per cent of labour) and 3 (50 per cent of labour) radio tape recorders. Similarly, India can produce 50 pens or 2 audio tape recorders or alternatively 25 pens (50 per cent of labour) and 1 audio tape recorder (50 per cent of labour). Now, we comes the of opportunity cost. By at trading these two countries, I e Japan and is the benefit of trading India can produce pens and tape recorders, what both the countries have foregone is the benefit of trading

If Japan produces only audio tape recorder and India produces only pens in which they have comparative advantage, Japan produces 6 audio tape recorders and India produces 50 pens. Both the countries get the advantage by doing so and by trading with each other. India can produce either 50 pens or 2 tape recorders for one day of labour.



It does mean that the cost of labour of 2 tape recorders is equal to that of 50 pens ($2=50$ or $1=25$). Thus, the cost of 1 tape recorder is 25 pens in India. Similarly, Japan can produce either 60 pens or 6 tape recorders for 1 day of labour. Therefore, the cost of 1 tape recorder in Japan is equal to ($6=60$ or $1=10$) 10 pens.

The price of audio tape recorder in India is 25 pens and it is 10 pens in Japan. Suppose, Japan offers 1 audio tape recorder for 17.5 pens, India would accept the offer and export 17.5 pens to Japan and get 1 audio tape recorder. This offer is beneficial to India as the price of 1 audio tape recorder is 25 pens in India. This offer is advantageous to Japan also as Japan gets 7.5 more pens for 1 tape recorder (price of 1 audio tape recorder 10 pens in Japan).

Gains from Trade Based on Comparative Cost Advantage		
	Prices with Trade	Prices without Trade
Japan	1 audio tape recorder = 10 pens	1 audio tape recorder = 17.5 pens
India	1 audio tape recorder 25 pens	1 audio tape recorder = 17.5 pens

Japan can get the benefit of more pens for 1 audio tape recorder ($17.5-10=7.5$ pens benefit), India also gets as India needs to spend only 17.5 pens to buy 1 tape recorder rather than 25 pens. It does mean that India saves ($25-17.5=7.5$) 7.5 pens in buying 1 tape recorder. India can buy 0.43 ($7.5/17.5=0.43$) more tape recorders with the benefit of 7.5 pens. It does mean that India can buy 1.43 (one for 17.5 pens and 0.43 for 7.5 pens) tape recorders for 25 pens. As such, Japan and India benefit from the trade. Thus, comparative cost advantage motivates the countries to trade with each other.

Implications of the Theory

- Efficient allocation of global resources
- Maximisation of global production at the least possible cost.
- Product prices become more or less equal among world markets.
- Demand for resources and products among world nations will be optimised.
- It is better for the countries to specialise in those products which they relatively do best and export them. It is better for the countries to buy other goods from other countries who are relatively better at producing them.

Comparative cost advantage theory is really an improvement over Adam Smith's theory of cost absolute advantage. This theory is not only an extension to the principles of division of labour and specialisation, but applies the opportunity cost concept. It is also argued that lower labour cost need not be a source of comparative advantage. However, Ricardo fails to consider the money value of cost of production. F.W. Taussig bridged this gap in comparative cost advantage theory.

IV. COMPARATIVE ADVANTAGE WITH MONEY

Modern economy is money economy and almost all the transactions take place in the form of money. Therefore, absolute differences in money prices determine international trade. According to FW Taussig, comparative differences in labour cost of commodities can be translated into absolute differences in prices without affecting the real exchange relations between products.

Table introduces money into discussion.

- Suppose the daily wage rate in Japan as on 1 March 2004 was Yen 360.
- Suppose the daily wage rate in India as on 1 March 2004 was ₹100.
- Suppose the exchange rate of Indian Rupee and Japanese Yen as on 1 March of 2004 was as follows: Rupee 2.00 Yen.

The Theory of Comparative Cost Advantage with Money: An Example				
	Cost of Goods in Japan		Cost of Goods in India	
	Japan Made	Indian Made	Japan Made	Indian Made
	₹	₹	₹	₹
Pens	6	4	3	2
Audio Tape Recorders	60	100	30	50

- Cost of producing pens in Japan = $360/60 = \text{Y } 6$
- Cost of producing audio tape recorder in Japan = $360/60 = \text{Y } 6$
- Cost of producing pens in India = $100/50 = ₹2$
- Cost of producing audio tape recorder in India = $100/2 = ₹50$

It is observed from Table that,

- In the absence of trade in Japan one pen costs Yen 6 (an equivalent of ₹ 3) and an audio tape recorder costs Yen 60 (an equivalent of ₹ 30) as on 1 March 2004.
- Similarly, in the absence of trade in India one pen costs ₹ 2 (an equivalent of Yen 4) and audio tape recorder costs ₹50 (an equivalent of Yen 100).
- If the two countries trade with each other, Indian made pen is cheaper in Japan, i.e., Yen 4 than Japan made pen, i.e., Yen 6
- Similarly, Japan made audio tape recorder is cheaper in India, i.e., ₹ 30 than Indian made audio tape recorder, ₹ 50

It is beneficial to both the countries to use Indian pens and Japanese audio tape recorders.

Japan has comparative cost advantage in producing audio tape recorders and India has comparative cost advantage in producing pens. Therefore, India can import audio tape recorders from Japan and Japan can import pens from Indian.

Criticism

- 1. Two Countries:** It is assumed that two countries participate in international trade. But, in reality more than two countries participate in international trade.
- 2. Transportation Cost:** It is criticised that the assumption of non-existence of transportation cost does not hold good as transportation cost is part of the process of global trade.
- 3. Two Products:** in reality, many products are involved in international trade. As such the assumption of existence of two products does not hold good.

- 4. Full Employment:** The assumption of full employment of resources is not a valid assumption as unemployment of resources is a normal feature in many countries.
 - 5. Economic Efficiency:** The goal of the nations in international trade is not necessarily economic efficiency. The other goals include helping the poor nations, trading with friendly nations, etc.
 - 6. Division of Gains:** Though the comparative cost advantage theory indicates that international trade provides gains to the trading nations, it does not provide the ratio at which the gains are shared between the trading nations.
 - 7. Mobility:** It is criticised that this theory does not consider the mobility of resources internationally. But, globalisation of economies provide for the free movement of all resources internationally.
 - 8. Services:** Comparative advantage theory deals with products, but not services. But, trading in services assumes significant share in global business, particularly in recent times.
- These criticisms led to the development of other theories on international trade. One among such theories is the relative factor endowments theory.

V. RELATIVE FACTOR ENDOWMENTS/HECKSCHER-OHLIN THEORY

In view of the criticism of cost against comparative advantage theory, the question pointed out by many was. How do the countries acquire comparative advantage? Eli Heckscher and Bertil Ohlin - Swedish economists-developed the theory of relative factor endowments to answer this question. Factor endowments are land, capital, natural resources, labour, climate, etc. The observations made by these two economists include:

- **Factor endowments vary among countries:** For example, the USA is rich in capital resources, India is rich in labour, Saudi Arabia is rich in oil resources, South Africa and Papua New Guinea have gold mines, etc. presents per capita GDP as relative factor endowment of selected countries as presented by International Monetary Fund.
- It is observed from Table 2.5 that Qatar had highest relative factor endowment in terms of per capita GDP (PPP) in the world in 2016 followed by Luxemburg, Macau, Singapore and Brunei. USA got 6th rank in terms of per capita GDP in 2016. India stood at 133rd position and its per capita GDP (PPP) in 2016 was only 3,843. Thus, India had relatively less endowments.
- According to these economists, if labour is available in abundance in relation to land and capital, in a country, the price of labour would be low and the price of land and capital would be high in that country. The vice versa is true in those countries where land and capital are available in abundance in relation to labour.
- These relative factor costs would lead countries to produce the products at low costs.
- Countries have comparative advantage based on the factors endowed and in turn the price of the factors. Countries acquire comparative advantage in those products for which the factors endowed by the country concerned are used as inputs.
- **For example,** India and China have comparative advantage in labour-intensive industry like textile and tobacco, Saudi Arabia has comparative advantage in oil. Therefore, countries export those goods in which they have comparative advantage due to factors endowed.
- Countries participate in international trade by exporting those products which they can produce at low cost consequent upon abundance of factors and import the other products which they can produce comparatively at high cost.

1. Land-labour Relationship: Countries where area of land available is less in relation to the people, go for multi-storey factories and produce lightweight products.

For example, clothing production in Hong Kong. Countries with large area of land in relation to population can go for sheep, wheat and other agricultural related products.

For example, Canada, Australia, India, etc.

2. Labour-capital Relationship: Countries where labour is abundant in relation to capital can be expected to export labour-intensive products, and vice versa is true in case of capital abundant countries. Thus, labour abundant countries acquire export competitiveness in products requiring large amounts of labour compared to capital.

For example, India has export competitiveness in textile garments while Iran has export competitiveness in handmade carpets. Japan has export competitive advantage in products requiring large amounts of capital relative to labour like computers, televisions, refrigerators, cars, etc. However, this generalisation has an exception.

3. Leontief Paradox: There are certain surprising aspects to the labour-capital relationship in international trade. Wassily Leontief observed that US exports are labour intensive compared with US imports. But, it is assumed that the USA has abundant capital relative to labour. Therefore, this surprise finding is known as the Leontief Paradox. This is because of variation in labour skills. Advanced countries have higher labour skills compared to developing countries. Therefore, advanced countries have competitive advantage in exporting products requiring higher labour skills while the developing countries have advantage in exporting products requiring less skilled labour.

4. Technological Complexities: Technological advancements made it possible to produce products in different methods. Canada produces wheat with more machines and India produces mostly with labour. In addition, industries locate different production processes in different countries in order to reduce cost of production.

This theory explains relative advantage of the countries based on the factor endowments. Thus, the theories discussed so far, are country based theories rather than firm based theories. Now, we shall discuss firm based theories. Firm based theories include: country similarity theory, product life cycle theory and global strategic rivalry theory.

B.

MODERN FIRM-BASED THEORIES

VI.

COUNTRY SIMILARITY THEORY

International trade takes place between two industries of two countries. But, international trade also takes place within each industry (i.e., intra-industry) between two countries. Intra-industry trade amounts to nearly 40 % of world trade. For example, Japan exports Toyota cars to Germany, whereas Germany exports BMW cars to Japan. This peculiar phenomenon happens due to the fact that Japan markets provide prestige and performance seeking automobile buyers to BMW cars, while German market provides quality conscious and value-oriented customers to Toyota cars.

1. Economic Similarity of Developed Countries

Steffan Linder a Swedish economist explained the phenomenon of intra-industry trade in 1961. According to Linder, the similarities in consumer preferences in the countries that are at the same stage of economic development provide the scope for intra-industry trade among countries.

For example, India and China are in the same stage of economic development. India provides market for low-cost motorcycles produced in China due to the price consciousness of the lower income group customers, whereas China's rich income group customers provide market for India's motorcycles due to their quality consciousness. Thus, this theory suggests that intra-industry trade takes place between the countries with similar levels of development.

According to this theory, the companies that develop new products for the domestic market, export the products to those countries that are at similar level of development after meeting the needs of the domestic market. Most part of the global trade takes place among developed countries as opposed to less developed countries as the former countries have acquired advantage and the latter countries have

natural advantage, particularly in agricultural products. The value of products produced through acquired advantage/technology is significantly higher than those produced through natural advantage. Advanced countries do also specialise in certain products to gain from acquired advantage though they have similarities.

For example, Switzerland specialises in producing pharmaceutical products, Germany specialises in machinery and equipment and Denmark in food products

However, developing countries mostly do not trade among themselves as the surplus of most of these countries would be raw materials and agricultural products and their requirements would be technology and high technology oriented products. As such, they export raw material to and import technology from developed countries.

For example, Vietnam, Papua New Guinea and Ethiopia export raw material, labour-intensive products and agricultural products to and import technology and finished products.

2. Similarity of Location

Countries prefer to export to the neighbouring countries in order to have the advantage of less transportation cost. For example, Papua New Guinea's majority of exports are to Australia and Finland is a major exporter to Russia due to less transportation costs. Similarly, some of the SAARC countries prefer to import from India.

3. Cultural Similarity

Countries prefer to export to those countries having similar culture. For example, exports and imports among European countries, between the USA and Canada, among Asian countries, and among Islamic countries. Similarly, trade among UK and its former colonies like India, Pakistan, South Africa, Australia and Papua New Guinea take place due to the similarities being ruled by UK. Nearly 48 per cent of foreign trade of Papua New Guinea was with Australia until 2009.

4. Similarity of Political and Economic Interests

Similar political interests, close political relations and economic interests enable the countries to enter into agreements for exports and imports. Countries prefer to trade with their politically friendly countries.

For example, India used to export to the former USSR and Pakistan prefers to export to USA. The enmity of the USA with Cuba resulted in the USA importing sugar from Mexico by abandoning sugar import from Cuba

VII.

PRODUCT LIFE CYCLE THEORY

The second firm based theory of international trade is the product life cycle theory. Raymond Vernon of the Harvard Business School developed the product life cycle theory. International product life cycle theory traces the roles of innovation, market expansion, comparative advantage and strategic response of global rivals in international manufacturing, trade and investment decisions.

International product life cycle consists of four stages, viz.

- New product introduction
- Growth
- Maturing product, and
- Decline

1. STAGE-1: NEW PRODUCT

Firms innovate new products based on needs and problems in the domestic country.

Examples:

- Photocopier and personal computer were innovated in the USA based on the perceived need.

- US firms became the leaders in the production of frozen food consequent upon the large kitchens, cheap electricity and more number of working women.
- French firms developed food packaging that would eliminate the need for refrigeration, consequent upon small freezer compartments.

A. Location of Innovation

Though the innovated product can be produced anywhere in the world and marketed in the domestic market, the firm mostly locates the manufacturing facilities in the domestic country to have immediate market feedback to be able to modify and develop the product accordingly and to save time and cost of transportation.

About 95 per cent of innovations (both products and technology) take place in industrially advanced countries. This is mostly due to severe competition, customer demands, availability of R & D facilities including scientists and engineers, financial ability of the firms, etc.

B. Significance of Innovation

Innovation is the prime source of competitive advantage. Leading firms innovate continuously and develop the products continuously in order to be on the forefront as innovations are imitated and copied the innovations can be in products, manufacturing process and marketing the product. Developing countries can have the utility of innovations by importing either technology or products from the R&D intensive countries.

During this stage, the firms sell most part of their products in domestic country and a limited part in other countries. Microsoft sold most part of its innovated software in the USA and the remaining part in various countries including India.

C. Labour as a Major Input: Production process at this stage is labour-intensive. This is because, the product in this stage is improved based on customer feedback. LG TV was modified more than ten times during this stage. The product is not yet standardised in this stage. Standard machinery and equipment to produce the product cannot be developed due to frequent changes in the product design. As such, production during this stage is labour-intensive. Even computer manufacturing was labour-intensive during this stage.

D. Feedback and Development: The firm's marketing executives have to continuously get the feedback from the customers and the production employees develop the product continuously.

2. STAGE-2: GROWTH

Increase in the sales of the new product attracts the competitors. At the same time, the increased awareness of the new product in various countries particularly in advanced countries increases the demand for the product. Added to this, further innovation in product, cost reduction, market process, etc., takes place resulting in increased capital intensity of the industry. These three factors influence either the innovator or competitor or both to produce in foreign countries in order to attain lower cost of production, lower transportation cost, better quality of the product, and the like Production in foreign countries mostly increases the sales in the foreign markets.

The competitors at this stage may concentrate mostly on the process technology rather than on product technology. Technology at this stage would be more of labour-intensive type. Thus, competitors start producing the product in various foreign markets at this stage. However, the original producing country would increase its exports to various countries though the competition has emerged in certain important foreign markets where local production has taken place.

3. STAGE-3: MATURITY

Worldwide production increases during this stage along with the demand for the product resulting in decline in exports. The increased competition results in increased product standardisation and cost

reduction. The producers start gaining the economies of scale reducing the cost of production per unit. The lower per unit cost of production results in exports to developing countries. At this stage, technology becomes standard. Therefore, the producers start locating their plants in developing countries in order to take the advantage of lower labour costs. This factor further reduces the cost of production per unit and increases the competition based on cost.

4. STAGE-4: DECLINE

Markets for the product at this stage concentrate in less developed countries as the customers in advanced countries shift their demand to further new products. Thus, most of the production plants at this stage locate in developing countries and exports decline considerably at this stage.

Even the original country may become net importer during this stage. Thus, the volume and direction of international trade come down considerably in this stage.

Suitability of the Theory: The tests conducted to evaluate this theory found that this theory was inconsistent for certain consumer durables, synthetic materials and electronics. However, some studies have found that the production movements did not take place as predicted in the Product Life Cycle (PLC) model.

Limitations of Product Life Cycle (PLC) Theory

- Production facilities do not move to foreign countries to achieve cost reductions due to short product life cycle consequent upon very rapid innovations.
- Cost reduction has little concern to the consumer in case of luxury products.
- Exports may not be in significant volume where cost of transportation is very high.
- Non-cost strategies like advertising may nullify the opportunity to move to foreign countries for cost minimisation.
- Requirement of specialised knowledge or expertise reduce the chances of locating production facilities in foreign countries. **For example**, the production of medical equipment could not be shifted from the USA and rotary printing process could not be shifted from Germany
- The rapid technological development may not shift the production to various foreign countries. **For example** : Microchip production which moved to various countries returned to the USA.
- The globalisation of world economies after 1990 enabled the transnational global companies to introduce innovations simultaneously in many countries where they operate.
- MNCs establish production process in many countries to take the advantage of raw materials, human resources, transportation cost, etc., but not based on the predictions of PLC model.

Various theories of international trade give us a general picture regarding the possibilities of international trade. But, the actual decision about international trade is taken by the individual companies. Now, we shall discuss why companies trade internationally.

VIII.

GLOBAL STRATEGIC RIVALRY THEORY

US airlines attracted both the domestic and international passengers through networking with other airlines, frequent flyer schemes, etc., after liberalisation in 1978. Consequently, European airlines lost significant business. KLM entered into strategic alliance with Northwest airlines, Jet Airways and a number of other airlines throughout the globe in order to recover the loss. Thus, MNCs acquire and develop competitive advantages in order to neutralise those of their rivals and/or to have leverage over their rivals in the process of globalising their operations.

Though the international trade theories, we have discussed so far, emphasise on the countries, international trade normally takes place between two companies of two countries. Further, international

trade takes place only when the companies have competitive advantage regardless of the countries.

Paul Krugman and Kelvin Lancaster have developed the view that firms struggle to acquire and develop some sustainable competitive advantage. This theory focuses on firms' strategic decisions to acquire and develop competitive advantage in order to compete internationally.

MNCs acquire and develop competitive advantage through a number of means. They are broadly categorised as follows

- Owning Intellectual Property Rights
- Investing in Research and Development
- Achieving Large-scale Economies
- Exploiting the Experience Curve

1. Owning Intellectual Property Rights: Firms which own an intellectual property right in the forms of patent, brand name, copyright and trademark acquire competitive advantage over their competitors.

For example, Kodak, LG, Coca-Cola, and the likes have competitive advantage over their competitors.

2. Investing in Research and Development: Investment in research and development would probably result in the development of new products, improvements to the existing products and development of new technologies, etc. These developments provide competitive advantage to the firms. Boeing spent US\$2 billion and developed the 747 jet and the US air spent US\$45 billion and developed the 747 aircraft. The firms that gain the competitive advantage through research and development have the first mover advantages.

3. Achieving Large-scale Economies: Companies with large-scale operations enjoy low cost of production/operations per unit. These companies may enjoy low-cost leadership.

4. Exploiting the Experience Curve: Production cost per unit tends to decline with the increase in the experience of the firm in manufacturing in case of certain industries. This is due to increase in employees' experience, expertise and skill. The companies gain global competitive advantage with the presence of experience curve. Samsung has the lowest cost advantage in manufacturing standardised semiconductor chips.

IX.

PORTER'S NATIONAL COMPETITIVE ADVANTAGE THEORY

Competitive Advantages: A recent study concluded that competitive superiority is derived from four factors, viz demand conditions, factor endowment, related and supporting industries and firm strategy, structure and rivalry. **Following Figure presents the determinants of global competitive advantage.**

This figure is also known as the Porter Diamond.

Should understand the combined effect of these factors on the development and continued existence of competitive advantages. All the four factors need not always be favourable for a company to get global supremacy. But, the interactive effect of these four factors need to be favourable if an industry/company in a country is to gain a global competitive advantage Now, we shall discuss the four elements of the diamond.

1. Factor Conditions: Factor conditions include factors of production. Hecksher-Ohlin theory deals with the classical factors, viz, land, labour, capital and organisation. Porter emphasises other factors like educational level of labour and quality of the country's infrastructure. Country's ability to compete globally depends upon the country's factor resources, viz., research, innovation and training. The USA has rich factor endowments and enjoys top position in world trade and world economy.

2. Demand Conditions: The existence of a large number of sophisticated domestic consumers who are economically able and willing to consume, create and improve the demand for various products in the country. Companies improve the existing products and develop new products to meet the existing and

new products. As such some of the processing domestic companies would be ahead of the international companies and export to other countries.



For example, Japanese companies developed camcorders, big screen TVs and VCRs better than the international companies and exported them to the European and North American countries after meeting the domestic demand.

3. Related and Supported Industries: The emergence and growth of an industry provide the scope for the development of suppliers of raw material, market intermediaries, financial companies, consulting agencies, ancillary industries, etc. These supporting service agencies compete among themselves leading to high input quality and lower prices. Availability of high-quality inputs at lower prices in the domestic country enhances competitive advantage of the firm internationally.

4. Firm Strategy, Structure and Rivalry: Firm continuously improve the quality, product design; invest in order to compete domestically. Firms also invest in human resource development, technology in the domestic market. These developments result in high-quality and lower prices in domestic country which are transferable to international markets. Intense competition for Japanese automobile manufacturers and electronics goods manufacturers led to their success in international markets. The same theory holds good in case of Indian garment manufacturers and US personal computer manufacturers.

CONCLUSION

Porter's theory blends the traditional country based theories and the firm based theories. Have to take all the theories into consideration to understand and explain the trade structure, direction and flows among countries. Firm based theories are more helpful rather than country based classical theories.

Strategic Advantages of Exports: Strategic advantages of exports include: Use of excess capacity, cost reduction per unit, greater profitability and risk spreading

- **Use of Excess Capacity:** Companies whose production capacity is more than the domestic demand tend to export, Companies in smaller countries have this advantage as the demand for the product in the domestic country would be less than the production capacity. Volvo could export from the small Swedish market than the General Motors of the USA.
- **Cost Reduction:** Research studies indicate that cost of production per unit can be reduced by 20 to 30 per cent when output is doubled. Cost reduction is due to coverage of fixed cost over a large

output, increasing efficiency due to experience gained, lower input rates due to purchases in larger quantities, etc. Experience curve also contributes for cost reduction.

Companies with lower cost advantage gain the opportunity to export.

For example, Matsushita and Toyota of Japan gained the opportunity to export based on the low cost of production.

- **Greater Profitability:** Producers may have the advantage of selling the product at profitable price in a foreign market. This situation happens due to the different stages of product life cycle in different countries. Mature or decline stage in domestic country and growth stage in foreign market enable the domestic company to export the product at a profitable price.
- **Risk Spreading:** Companies can spread the risk of business by exporting the product to a member of countries. Donaldson company a US manufacturer of filters, air cleaners, etc., could achieve record earnings in 1991 despite recession in domestic market.

X.

POINTS TO BE REMEMBERED

- International trade takes place due to differences in opportunity costs.
- According to mercantilism, countries should export more than they import and receive the difference in gold.
- Decay of gold standard reduced the validity of this theory.
- According to Adam Smith, free trade enables a country to produce a variety of goods and services.
- Every country should specialise in producing those products at the cost less than that of other countries.
- Natural advantage is due to climatic conditions, natural resources, etc.
- Acquired advantage is due to technology and skill development.
- Country should produce and export those products for which it is relatively more productive than other countries
- and import other products
- Factor endowments are: land, capital, natural resources, labour, climate, etc., and they vary among countries.
- International trade takes place among the countries that are at the same stage of economic development.
- PLC theory provides inputs for manufacturing, trade and investment decisions.
- International trade takes place between/among companies based on relative competitive advantage, but not countries competitive advantage.
- Companies get competitive superiority from: demand conditions, factor endowment, related and supporting industries, firm strategy, structure and rivalry.